

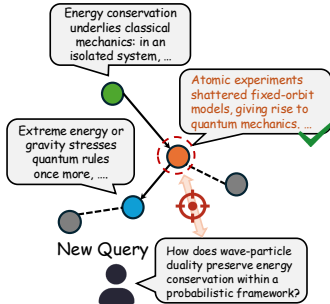
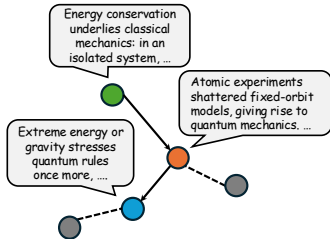
Energy conservation underlies classical mechanics: in an isolated system, total energy merely toggles between potential, kinetic, and other forms; a swinging pendulum illustrates this. Atomic experiments shattered fixed-orbit models, giving rise to quantum mechanics. Wave-particle duality preserves energy bookkeeping probabilistically, explaining atomic spectra and enabling semiconductor technology. Extreme energy or gravity stresses quantum rules once more, demanding relativity and yielding quantum field theory. Electroweak and strong forces now fit this scheme, while string and loop quantum gravity aim to complete energy-conserving unification for all four interactions.

Historical Memory

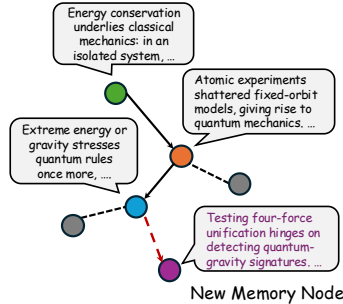


Memory Graph

Memory Organization



Memory Retrieval



Memory Maintenance